Automated Network Request Management

Phase 3: Project Planning Phase — Implementation Roadmap, WBS, RACI, Risk Matrix & PDI Replication Guide

1. Implementation Roadmap & 5-Phase Work Breakdown Structure (WBS)

To ensure a seamless, error-free transition from architectural design to technical execution, the deployment was structured into a 5-Phase Work Breakdown Structure (WBS). Each milestone represents an immutable validation gate:

Phase / Milestone	Primary Deliverables	Validation Checkpoint	Status
Phase 1: Catalog Creation	• Create 'Network Request' Catalog Item • Define 'Request Information' Variable Set • Build UI variables & choice options	Item appears in Service Catalog under Standard Changes / Network Services	COMPLETED
Phase 2: Form & Data Setup	• Build custom table u_database_tables • Setup auto-numbering padded format • Create 'Approval Request' Relationship	Table stores records with auto-generated IDs; Related list displays on form	COMPLETED
Phase 3: Flow Automation	• Configure Flow Designer 'Network Requests' • Implement Action: Get Catalog Variables • Configure Record Creation, Approvals, & Emails	Flow executes end-to-end upon catalog item ordering in test harness	COMPLETED
Phase 4: Security & Testing	• Verify table ACLs for ITIL & End-Users • Conduct end-to-end Service Portal tests • Verify email logs (sys_email) & audit trail	100% test case pass rate with clean Flow execution logs	COMPLETED
Phase 5: Deployment & Handover	• Production readiness sign-off • Admin & Fulfiller operational guides • Scope future Integration Hub / Ansible roadmap	Operational documentation and sign-off approved by IT Leads	COMPLETED

2. Responsibility Assignment Matrix (RACI Framework)

The RACI framework clarifies organizational accountability across each project phase:

Project Activity	Requester	IT Admin / Dev	Network Team	Approver
Requirements & Ideation	Consulted (C)	Responsible (R)	Consulted (C)	Accountable (A)
Table Schema & Relationships	Informed (I)	Responsible (R)	Consulted (C)	Accountable (A)
Catalog Form & UI Policies	Consulted (C)	Responsible (R)	Consulted (C)	Informed (I)
Flow Designer Automation	Informed (I)	Responsible (R)	Consulted (C)	Accountable (A)
Service Portal User Acceptance	Responsible (R)	Accountable (A)	Consulted (C)	Informed (I)
Approval & Governance Verification	Informed (I)	Responsible (R)	Informed (I)	Accountable (A)
System Logs & Email Verification	Informed (I)	Responsible (R)	Informed (I)	Informed (I)

3. Risk Assessment & Threat Modeling Matrix

Identified Risk	Severity	Impact Description	Mitigation Strategy
Variable-to-Table Data Misalignment	High	Catalog inputs fail to map cleanly to custom table fields during record generation.	Enforce strict variable naming conventions and test data pill mapping in Flow Designer action inputs.
Approval Routing Timeouts	Medium	Managers fail to act on requests, causing breach of the 48-hour fulfillment SLA.	Configure automatic reminder notifications and escalation paths in Flow Designer logic.
ACL Permission Lockout	High	End users or fulfillers unable to view or edit required fields on u_database_tables.	Retain baseline platform table ACLs with targeted role assignments (itil and snc_internal).
Email Notification Suppressions	Medium	Outbound emails held in queue or filtered by corporate spam filters.	Verify sys_email outbox logs and validate standard SMTP notification headers.

4. Quality Assurance & Test Strategy Framework

A rigorous multi-tiered testing plan was designed to guarantee system integrity prior to production rollout:

- **Client-Side Validation Testing:** Verify that mandatory fields block form submission when empty, and dynamic UI policies accurately toggle the 'Existing ID' field based on connection type.
- **Flow Execution Traceability:** Review Flow Designer Execution Details step-by-step to verify that variables are extracted, records are inserted, and approval states trigger the correct IF/ELSE logic branch.
- **Audit & Email Log Inspection:** Check sys_email table to verify that notification triggers fired, message delivery status was marked 'send-ready', and recipient addresses matched the requester.
- **Role-Based Impersonation:** Impersonate standard non-admin users to guarantee that portal access, approval buttons, and table security boundaries comply with enterprise least-privilege standards.

5. Personal Developer Instance (PDI) Replication Manual

To reproduce this complete automated solution in any standard ServiceNow Personal Developer Instance (PDI), execute the following step-by-step instructions in exact sequence:

Step	Module / Navigation	Action & Configuration Details
1	System Definition > Tables	Click **New**. Label: Database Tables, Name: u_database_tables. Add columns (Number, Database Number, Assignment to, Mobile Number, Total Amount, Mode of Payment, Type of Connection, Customer Address, Approval Status, Requested For).
2	System Definition > Relationships	Click **New**. Name: Approval Request. Applies to: u_database_tables. Queries from: sysapproval_approver. Script: current.addQuery('sysapproval', parent.sys_id);. Save and add Related List to form.
3	Service Catalog > Catalog Items	Click **New**. Name: Network Request. Catalogs: Service Catalog. Category: Standard Changes. Short description: Network request Management.

4	Service Catalog > Variable Sets	Create Single-Row Variable Set: Request Information. Add variables (Opened on behalf of, User Name, Mobile Number, Type of Connection [Choices: NEW, EXISTING], Existing ID, Total Amount, Mode of Payment, Email Id, Phone Number).
5	Catalog Item > Catalog UI Policies	Click **New**. Condition: Type of Connection is EXISTING. Save. In Related List 'Catalog UI Policy Actions', add Existing ID with **Visible: True**. Set Reverse if false = True.
6	Process Automation > Flow Designer	Click **New Flow**: Network Requests. Trigger: Service Catalog. Add Action 1: *Get Catalog Variables*. Action 2: *Create Record* in u_database_tables. Action 3: *Send Email*. Action 4: *Ask for Approval*. Flow Logic 5: *If Approved* → Update Record (Approval Status = approved). Flow Logic 6: *Else* → Update Record (Approval Status = rejected). **Save & Activate**.
7	Service Portal (/sp)	Navigate to https://<instance>.service-now.com/sp. Search 'network request'. Fill form, verify dynamic visibility of Existing ID, and submit. Confirm REQ creation, approval generation, and email trigger in sys_email.

Phase 3 Sign-Off: With the WBS, RACI assignments, risk controls, test strategy, and PDI replication manual defined, the engineering team proceeds to Phase 4 for technical development, execution testing, and audit validation.